

PROJECT DEVELOPMENT DOCUMENTATION PACK

Integrated Construction & Manufacturing Workforce SaaS
ASP.NET Core MVC • SQL Server • Android • iOS • Multi-tenant SaaS

PROJECT BASELINE	INITIAL RECOMMENDATION
Engagement	Custom development for a paying client
Functional scope	Safety & EHS, Quality/Snagging, ePTW, Estimation/Quotation, Workforce
Architecture	ASP.NET Core MVC modular monolith, EF Core, SQL Server, REST APIs
Mobile	Shared cross-platform Android/iOS application
AI	Phase 2; not in initial release
Commercial reference	■42,00,000 development fee + applicable GST, per separate quotation

Purpose: This pack identifies the documents required to plan, build, test, deploy and hand over the product. It includes an initial documentation baseline and example requirements/diagrams. It is a starter pack, not a final approved specification.

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1. Document Register — What Is Needed?

ID	Document	Purpose / key contents	Priority
D01	Business Requirements Document (BRD)	Business goals, target users, problems, scope, business outcomes and success metrics.	Must have
D02	Product Requirements Document (PRD)	Product vision, personas, modules, user journeys, MVP boundaries and roadmap.	Must have
D03	Software Requirements Specification (SRS)	System behavior, functional/non-functional requirements, constraints and interfaces.	Must have
D04	Functional Requirements & Use Cases	Detailed module workflows, roles, alternate paths, validations and business rules.	Must have
D05	UI/UX Specification	Sitemap, navigation, wireframes, screen inventory, responsive/mobile behavior and design system.	Must have
D06	Solution Architecture Document (SAD)	Logical/deployment architecture, component boundaries, security, environments and technology choices.	Must have
D07	Database Design / ERD	Entities, fields, keys, relationships, tenant strategy, indexes, retention and migration plan.	Must have
D08	API Specification	Endpoints, request/response schemas, authorization, error formats, versioning and examples.	Must have
D09	Security & Privacy Plan	Threat model, access control, tenant isolation, encryption, audit, retention and incident response.	Must have
D10	Test Strategy & Test Cases	Unit/integration/UI tests, device coverage, UAT scenarios, defect severity and exit criteria.	Must have
D11	Project Plan / Delivery Schedule	Milestones, dependencies, owners, risks, assumptions, change control and acceptance gates.	Must have
D12	DevOps / Release Plan	CI/CD, branching, environments, secrets, database migrations, rollback, monitoring and backups.	Must have
D13	Deployment & Operations Runbook	Provisioning, deployment steps, restore drills, alert handling, support escalation and disaster recovery.	Must have
D14	User/Admin Guides & Training	Role-based user guides, admin setup, training agenda and handover checklist.	Must have
D15	Data Migration Plan	Source data inventory, mapping, cleansing, trial migration, reconciliation and rollback.	If migration
D16	Integration Specifications	External API contracts, credentials, rate limits, retries, error handling and ownership.	If integrations
D17	AI Feature Specification	Model/provider, allowed data, evaluation, human review, safety controls and costs.	Phase 2
D18	SLA / Support & Maintenance	Support hours, severity levels, response targets, backups, patching, uptime and exclusions.	Must have
D19	Commercial SOW / Contract	Scope, fees, milestones, GST, IP, warranty, change requests and termination terms.	Must have
D20	Risk Register & Decision Log	Open risks, owners, mitigation, assumptions and recorded architecture/product decisions.	Must have

Recommended approval order: BRD → PRD → SRS/FRD → UI/UX → Architecture/ERD/API → test and release plans → signed Statement of Work.

2. Product Scope and MVP Boundary

Product goal

Deliver one connected operational platform for construction and industrial SMEs. Shared records should connect organizations, sites, projects, workers, contractors, permits, inspections, incidents, estimates and corrective actions.

Phase 1 modules

- Core SaaS: tenants, users, roles, sites, projects, audit trail, document uploads and notifications.
- Safety & EHS: incidents, near misses, observations, toolbox talks and corrective actions.
- Quality: inspection templates, checklists, defects/snags, assignments, reinspection and closure.
- ePTW: permit templates, risk assessment, approval workflow, issue/suspend/resume/close and history.
- Estimation: customer records, BOQ/BOM, cost lines, margin/tax configuration, estimate revisions and quotation PDFs.
- Workforce: employee/contractor onboarding, certification expiry, shifts, attendance and site allocation.
- Mobile: attendance, photos, inspections, toolbox acknowledgement, safety reports and assigned actions.
- Reporting: operational dashboards and exportable standard reports.

Explicitly out of Phase 1

- Generative AI, computer vision and predictive analytics; reserve for Phase 2.
- Full payroll/statutory compliance, ERP/accounting integrations not listed in the SOW, BIM/CAD takeoff.
- Full offline-first sync with conflict resolution unless separately approved.
- Separate native mobile codebases; use one cross-platform codebase.

3. Stakeholders, Roles and Permissions

Role	Primary permissions (subject to tenant and site scope)
Platform Super Admin	Manage tenants, platform configuration, subscriptions and platform health; no routine business-data access unless authorized.
Company Admin	Manage company settings, users, sites, role assignments and approval configuration.
Project Manager	Manage assigned projects, estimates, workforce allocation, reports and project actions.
Safety Officer	Manage incidents, toolbox talks, safety actions, risk assessments and permit reviews/approvals assigned to them.
QC Inspector	Execute inspections, record defects, assign corrective actions and verify closure.
Supervisor / Foreman	Record attendance, toolbox participation, observations and work progress for authorized sites.
Worker	View assigned tasks, submit attendance/observations, acknowledge talks and participate in authorized permits.
Contractor Representative	Maintain contractor documents, worker lists, certifications and assigned work records.

Permission rule: enforce authorization on the server for every request. Hiding a button in the UI is not an access-control mechanism.

4. Functional Requirements — Initial Baseline

ID	Requirement	Acceptance example
AUTH-01	Users can sign in, sign out and reset passwords.	Valid credentials sign in; invalid credentials are rejected; reset flow does not reveal whether an email exists.
TEN-01	Every tenant-owned record is scoped to an organization.	A user from Tenant A cannot read or modify Tenant B records through UI, API, exports or file URLs.
ORG-01	Company Admin can manage sites and projects.	Required fields are validated; duplicate identifiers are prevented within the tenant.
EHS-01	Authorized users can report incidents and near misses with evidence.	Record captures site, time, category, description, severity and optional attachments; action is audited.
EHS-02	Safety actions can be assigned and tracked.	Assignee, due date, status and closure evidence are stored; overdue actions appear in reports.
QC-01	Inspectors can execute configured inspection checklists.	Each checklist item records result, comments and evidence where required.
QC-02	Failed inspection items can create snags.	Snag links back to inspection/item, has owner and due date, and requires verification before closure when configured.
PTW-01	Authorized workers can submit permit requests.	Permit has type, site, work location, time window, hazards, controls and requested participants.
PTW-02	Approvals follow configured rules and are auditable.	Only authorized approvers can approve; all decisions record actor, timestamp and comments.
PTW-03	High-risk permit cannot be issued before required approvals.	System blocks issue when mandatory approvals/checks are missing.
EST-01	Users can create estimates from BOQ/BOM lines.	Quantity × rate and configured cost components calculate correctly; rounding rules are documented.
EST-02	Approved estimates generate branded quotations.	PDF includes client, line items, totals, validity, terms and version/reference number.
WF-01	Company Admin can onboard employees and contractors.	Required identity/document fields validate; certification expiry is tracked.
WF-02	Authorized workers/supervisors can record attendance.	Check-in/out times are stored; duplicate or invalid sequences are rejected or flagged.
NOT-01	System sends configured workflow notifications.	Notification is created for the correct tenant/recipient; failures are logged and retry policy is applied.
AUD-01	Material changes to sensitive records are audited.	Audit record includes actor, action, timestamp, record reference and relevant before/after values.
MOB-01	Mobile users can submit supported field workflows.	Mobile submissions validate on server; upload failures are recoverable; duplicate retries do not create duplicate records.
RPT-01	Authorized users can view standard dashboards/reports.	Totals respect tenant/site permissions and match the underlying filtered records.

5. Non-Functional Requirements (Initial Targets)

Area	Proposed requirement
Performance	Target p95 under 500 ms for common API reads under an agreed representative load; dashboard under 3 seconds for typical SME datasets. Validate with performance tests.
Availability	Initial target 99.5% monthly availability, excluding agreed maintenance and third-party outages; define measurement and remedies in SLA.
Security	HTTPS, secure secrets, least privilege, server-side authorization, secure upload validation, rate limiting and OWASP-aligned testing.
Tenant isolation	Tenant context resolved server-side from authenticated identity; no client-supplied TenantId is trusted without validation.
Privacy	Collect only required personal data; define purpose, retention, export/deletion process and client responsibilities.
Auditability	Record key workflow decisions and security-sensitive changes with actor and timestamp; restrict audit-log modification.
Accessibility/usability	Responsive layouts, keyboard-friendly controls and clear validation/error messages for agreed critical workflows.
Recovery	Define backup frequency, retention, Recovery Point Objective (RPO) and Recovery Time Objective (RTO) with the client.
Maintainability	Modular monolith, code review, automated tests, API versioning where needed and documented database migrations.
Mobile	Support agreed Android/iOS versions and device matrix; secure token storage; handle weak connectivity and retry safely.
Observability	Centralized application logs, health checks, error tracking and actionable alerts without logging secrets or sensitive payloads.

6. Proposed Solution Architecture

Web Browser / Mobile App

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| HTTPS
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ASP.NET Core MVC + REST API

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Application Use Cases / Validation / Authorization

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Safety/EHS Quality/QC ePTW

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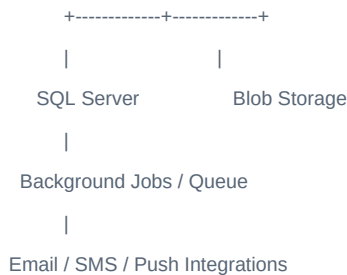
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Estimation Workforce Notifications

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Domain / Business Rules

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Infrastructure / EF Core

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AI Gateway / LLM / Vision / RAG: Phase 2, behind policy and human-approval controls.

Implementation recommendation: start with a modular monolith. Keep module boundaries and API contracts clear; introduce separate services only when scale, team ownership or isolation requirements justify the operational cost.

7. Core Data Model — Initial ERD Outline

Entity	Important relationships / fields
Organization	Tenant root; name, status, configuration, subscription reference.
User / Role / UserRole	Identity, role assignments, tenant membership and site/project scope.
Site / Project	Belongs to Organization; project belongs to site or organization-defined site structure.
Employee / Contractor / Certification	Belongs to tenant; worker may be allocated to site/project; certifications have expiry and evidence.
Shift / AttendanceRecord	Tenant + worker + site + shift; check-in/out, source, correction status and approver.
Incident / SafetyAction / ToolboxTalk	Tenant + site/project; reporter, category, severity, action owner, due date and closure evidence.
Inspection / InspectionItem / Snag	Inspection template/run, result items, defect owner, due date, verification and closure.
Permit / PermitApproval / RiskAssessment	Permit type, site/project, time window, participants, risk controls, approval state and audit history.
Estimate / EstimateItem / Quote	Customer, revision, BOQ/BOM lines, quantities, units, rates, cost components, tax and approval state.
Attachment / Notification / AuditLog	Tenant-scoped metadata, secure object-storage key, delivery status and immutable-style audit history.

Design rules: use primary keys, tenant foreign keys, created/updated timestamps, concurrency tokens where needed, soft deletion only where justified, and indexes for common tenant/site/status/date filters. Avoid storing uploaded file bytes directly in relational tables.

8. API Specification — Starter Endpoints

Area	Example endpoints
Identity	POST /api/v1/auth/login • POST /api/v1/auth/refresh • POST /api/v1/auth/password-reset
Sites / Projects	GET/POST /api/v1/sites • GET/POST /api/v1/projects • GET /api/v1/projects/{id}
Safety	POST /api/v1/incidents • GET /api/v1/incidents • POST /api/v1/incidents/{id}/actions
Quality	POST /api/v1/inspections • POST /api/v1/inspections/{id}/complete • POST /api/v1/snags • POST /api/v1/snags/{id}/verify
Permits	POST /api/v1/permits • POST /api/v1/permits/{id}/submit • POST /api/v1/permits/{id}/approve • POST /api/v1/permits/{id}/issue • POST /api/v1/permits/{id}/close
Estimation	POST /api/v1/estimates • POST /api/v1/estimates/{id}/approve • POST /api/v1/quotes
Workforce	POST /api/v1/workers • POST /api/v1/attendance/check-in • POST /api/v1/attendance/check-out
Reporting	GET /api/v1/reports/operations • GET /api/v1/reports/compliance

API standards to document: authentication, authorization, tenant resolution, pagination, filtering, validation errors, idempotency for retries, file upload limits, rate limits, versioning and OpenAPI/Swagger.

9. Test Strategy & Acceptance

- Unit tests: business rules, cost calculations, permit state transitions and authorization policies.
- Integration tests: SQL persistence, tenant filters, file storage, notification delivery and migrations.
- API tests: valid/invalid payloads, permission boundaries, cross-tenant access attempts, pagination and error formats.
- UI tests: primary journeys for each role and responsive layout checks.
- Mobile tests: supported OS/device matrix, camera/photo permissions, interrupted uploads, retries and secure session behavior.
- Security tests: authentication/session handling, IDOR/BOLA checks, file upload validation, secrets handling and dependency vulnerabilities.
- UAT: client-nominated users execute signed scenarios; record pass/fail, defects, owner and retest result.
- Release gate: no open critical/high defects; agreed medium/low defects documented with accepted workarounds or planned dates.

Example UAT scenario: A worker with an expired mandatory certification requests a high-risk permit. The system must flag the expired certification and prevent permit issue if that certification is configured as mandatory. The system records the reason and notifies the designated owner.

10. Project Plan & Milestones

Phase	Indicative schedule	Exit criteria
Discovery / BRD / PRD	Weeks 1–3	Scope, personas, user journeys, assumptions and MVP signed off.
Architecture / UX / ERD / API	Weeks 2–5	Architecture, initial schema, key screens and API conventions approved.
Core platform	Weeks 4–7	Tenant isolation, authentication, roles, sites and projects tested.
Module implementation	Weeks 8–15	Five module workflows delivered to agreed feature acceptance.
Mobile implementation	Weeks 10–18	Agreed mobile journeys pass device and API tests.
Integration / QA / UAT	Weeks 19–22	Critical defects resolved; client UAT completed.
Deployment / handover	Weeks 23–24	Production release, runbook, training and handover completed.
Contingency / stabilization	Weeks 25–28	Agreed remediation and external approval buffer.

Indicative payment schedule from the commercial quotation: 20% kickoff; 15% architecture/UI approval; 25% core platform/modules; 20% mobile/integrations; 15% UAT/deployment/handover; 5% warranty completion. Align the final contract with objective deliverables and review windows.

11. Security, Privacy & Operational Controls

- Use ASP.NET Core Identity or an equivalent approved identity provider; MFA for privileged roles where feasible.
- Resolve tenant context from authenticated membership and validate tenant/site/project authorization server-side on every operation.
- Store secrets in a managed secret store; never commit credentials or log access tokens.
- Use HTTPS and managed encryption; define database backup encryption and retention.
- Validate file extensions, MIME types, size, filenames and access permissions; use malware scanning where appropriate.
- Log security-relevant events while avoiding passwords, tokens and unnecessary personal data.
- Define data retention, export, deletion, breach escalation and client/vendor responsibilities before production.
- Review applicable Indian data protection requirements and contractual obligations with the client/legal adviser.

12. Deployment & Operations Checklist

- Separate development, test/UAT and production environments.
- CI/CD pipeline with code review, automated tests, dependency scanning and controlled approvals.

- Environment-specific configuration and secret management.
- Database migration review, backup before release and rollback plan.
- Health checks, centralized logs, alerts and error tracking.
- Scheduled backups and documented restore tests; agreed RPO/RTO.
- Production access restricted to named authorized personnel; access reviewed periodically.
- Mobile release signing keys and app-store accounts managed by the client.
- Runbook covers deployment, incident response, restore, certificate renewal and support escalation.

13. AI Phase 2 Readiness

AI is excluded from the initial release, but Phase 1 should store structured records and attachments consistently so later AI capabilities can be added without redesigning the platform.

- Potential features: safety/incident drafting, inspection photo assistance, estimate extraction and cross-module operational Q&A.;
- Define approved data sources, data access boundaries, tenant isolation and model/provider retention settings.
- Evaluate output quality using representative client examples; show uncertainty and preserve source references where practical.
- Require human review for safety-critical recommendations; never let AI independently approve high-risk permits.
- Budget separately for model/API usage, evaluation, monitoring, prompt-injection defenses and ongoing quality reviews.

14. Risks, Assumptions & Decisions to Confirm

Item	Assumption / decision required
Mobile approach	Cross-platform app; confirm framework and supported OS/device versions.
Offline use	Basic caching only in baseline; decide whether full offline workflows are mandatory.
Tenancy	Shared database with TenantId and server-side enforcement initially; confirm enterprise isolation needs.
Integrations	Only named integrations included; obtain API documentation and credentials early.
Data migration	No large historical migration; define source formats and volume if needed.
Regulatory workflows	Client supplies approved EHS procedures, permit templates, retention rules and required sign-off matrix.
Hosting / region	Client approves cloud provider, region, data residency and account ownership.
Acceptance	Sign testable acceptance criteria, review windows and defect severity definitions.
Scope / schedule	The 24–28 week timeline assumes timely decisions, stable scope and client availability.

15. Sign-Off Checklist

- BRD, PRD, SRS and FRD approved by business owner.
- UI screen inventory and mobile journeys approved.
- Architecture, ERD, tenant-isolation approach and API conventions approved.
- Security/privacy responsibilities and retention requirements agreed.
- Named client UAT representatives and acceptance criteria agreed.
- Deployment ownership, cloud budget, backup/RPO/RTO and support SLA agreed.
- Commercial SOW, GST treatment, payment milestones, IP terms and change-control process signed.

Prepared as a project-start documentation baseline. All requirements, timelines, architecture decisions, performance targets and commercial assumptions must be reviewed and approved by the client and delivery team before implementation.